Week 12 web page: delivery guide

Instructor notes for Week12_MixedModels.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week12_MixedModels.html. Script: `CODE_Week12_MixedModels.R` (same seeds, same numbers; run its first section before any later one).
- Veterans Day: no class Wed., Nov. 11. The week is Monday plus Tuesday's lab period.
- Monday has **no WOD** (the lecture is full). The required take-home WOD *Formula to Model* goes out Monday, due Mon., Nov. 16; budget 5-10 min that day to go over it. Key: `wod/answer_keys/`.
- Tuesday: ~60 min of lecture with laptops open (students run the Stan model), then an open final-project work session. No lab Act or key.
- Two simulated data sets; say so up front.
  - Pinyon chronologies (seed 21; 4 stands × 5 trees × 25 yr): truth stand SD 0.12, tree SD 0.10, precipitation slope 0.025 mm/cm, AR(1) $\rho = 0.6$, **no trend**.
  - Drought sensitivity (seed 12): 12 stands, records of 30, 24, 18, 12, 8, 5 years (two stands each), 194 stand-years. Precipitation in 10-cm units centered at 20 cm. Truth: $\mu_a = 0.9$, $\mu_b = 0.25$, $\tau_a = 0.15$, $\tau_b = 0.10$, $\rho = -0.6$, $\sigma = 0.12$.
- Questions on the page have no answers. *Land:* is where discussion should end up.

Part 1: Autocorrelation and mixed models in R (Monday, ~45 min)

Sections 1-2 are the Week 11 material that wasn't reached on Nov. 4; the Week 11 page now points here.

Big picture (~3 min)

- MODELS and FIT boxes. Recap in one breath: random intercept → block-diagonal V. Today: correlation through time, then groups that differ in slope, then the software.

AR(1) (~10 min)

- Management question first: is growth declining? `lm()`: trend −0.0055 mm/yr, SE 0.0026, $p = 0.045$. Stand A's lag-1 residual ACF (0.33) is inside the ±0.39 bounds: not detected ≠ absent. All four stands are positive (0.33, 0.33, 0.48, 0.41).

- Derive $\text{Var}(\eta_t) = \sigma^2/(1-\rho^2)$ on the board (two lines). Notation follows Hyndman & Athanasopoulos: η_t is the autocorrelated error, ε_t the independent innovation; `nlme` prints ρ as `Phi`. AR(1) fit: $\hat{\rho} = 0.37$; trend -0.0047, SE 0.0034, $p = 0.17$. `gls()` matches. Then tell them the truth has no trend.
- Widget (`cov_structures.html`): switch to AR(1), default ρ 0.6 (lag-1 0.60, lag-5 0.08). Ask students to predict the series at $\rho = 0.9$ before moving the slider: it wanders like a trend (lag-5 still 0.59). Then $\rho = -0.7$: zigzag, checkerboard heatmap.
- "With ρ near 0.9, what does a single AR(1) series look like? How could you tell it apart from a slow trend?" *Land:* long runs above and below the mean that look like trends. From one short series you often can't; you need a longer record, replicate series, or a mechanism. This is the fan in section 2.
- *If time:* AR errors vs. lagged response (`#lagged-response`). Board the substitution. Stand A: plain lagged fit $\hat{\gamma} = 0.02$ (truth $\rho = 0.6$); with last year's precip $\hat{\gamma} = 0.33$ and lagged precip -0.0089 vs. $-\hat{\gamma}\hat{\beta} = -0.0070$. Point: choose the form by mechanism (does the response itself carry over?).

Why harder (~7 min)

- Three points, one line each: cost (about 150 ms per MVN evaluation at $n = 2000$ on the render machine), awkward surfaces, constraints.
- Random-intercept surface: σ_b about 0.08 to 0.35; σ much tighter.
- Fan: near $\rho = 0$ the trend runs about -0.0074 to -0.0039 (excludes 0); at $\hat{\rho} = 0.37$ it runs -0.0113 to 0.0017.
- **Behind?** Skip the timing; show the two surfaces.

Random slopes (~8 min)

- Show the two-panel figure. Right panel: low-intercept stands have steep slopes. Ask what ecology would produce that (low elevation, closer to the water limit).
- Write the model and G on the board; count the six parameters. The 24 stand deviations are integrated out, not estimated.
- $V_j = Z_j G Z_j^T + \sigma^2 I$: one block per stand. Don't expand $\text{Var}(y)$ as a function of x (the take-home WOD asks for it).
- Formula table: `(1 | stand/tree)` expands to `(1 | stand) + (1 | stand:tree)`. Don't count parameters for them out loud beyond the table (the WOD uses different formulas).

Mixed models in R (~10 min)

- `lmer()`: $\hat{\mu}_a = 0.840$, $\hat{\mu}_b = 0.258$, $\hat{\tau}_a = 0.131$, $\hat{\tau}_b = 0.092$, $\hat{\rho} = -0.57$, $\hat{\sigma} = 0.118$. AIC -180.9 (random intercept) vs. -216.9 (random slope), a drop of 36.0.
- If asked about an LRT: $\chi^2 = 40.0$ on 2 df, $p \approx 2 \times 10^{-9}$; conservative because $\tau_b = 0$ is on the boundary (same issue as Poisson vs. negative binomial).
- The comparison table: `optim()`, `lmer()`, `lme()`, and `glmmTMB()` all give log-likelihood 114.4497. The point: the packages fit the likelihood we could write ourselves, faster and with better parameterizations.

- Package bullets: glmmTMB as the practical default (one syntax for non-Normal families, dispersion, AR(1) and spatial structures); lme4 for small-sample inference on Normal models (REML, `lmerTest`, profile intervals); nlme for the richest Normal residual structures. Point students to the reference handout `NOTES_Reference_MixedModelSyntax.pdf` (linked on the page).
- **Behind?** Show only the `lmer()` fits and the comparison table; skip the `optim()` code walk-through.

Discussion (~5 min; pick one or two)

- Q1 (trend SE 0.0026 to 0.0034; precipitation SE 0.0026 to 0.0020) *Land:* year is itself perfectly autocorrelated, so runs of correlated errors line up with it and mimic a trend; precipitation jumps around, so correlated errors don't line up with it. Errors matter most for predictors that change slowly in time or space.
- Q2 (four stands: fixed or random?) *Land:* the random-intercept model answers "how variable are stands in general" and can predict a new stand, but σ_b from four stands is weak (wide surface). The fixed model answers "how do these four stands differ." Choose by the question; with few groups, say so and report the uncertainty.
- Q3 (four short-record stands: $\hat{\tau}_a = 0.084$, $\hat{\tau}_b = 0.063$, $\hat{\rho} = -1.00$, singular fit) *Land:* no; 26 stand-years in 4 stands can't support three variance parameters, and $\hat{\rho} = -1$ is the boundary (the fit collapsed to one dimension). Report it as singular; simplify to `(precip10 || stand)` or a random intercept, get more stands, or use a prior (Tuesday).

Bonuses (only if time): AR(1) by GLS (closed-form $\hat{\beta}(\rho)$ and $\hat{\sigma}^2(\rho)$, then `optimize()` over ρ; matches `optim()` and `gls()`, $\hat{\rho} = 0.367$, logLik 27.037; profile interval for ρ −0.03 to 0.75, so neither 0 nor the true 0.6 is ruled out; `gls()` reports $\sigma/\sqrt{1-\rho^2}$, not the innovation SD); random intercept + AR(1) ($\hat{\rho} = 0.51$, log-likelihood 82.28, matches `lme()`); Poisson GLMM by hand (`integrate()`: 1.551, 0.382, σ_b 0.542; `glmmTMB()` 1.548, 0.382, 0.542).

Part 2: Bayesian hierarchical models (Tuesday lab lecture, ~60 min)

Before class: remind students (Canvas) to bring a laptop with `cmdstanr`, `brms`, and `lme4` installed. In class they run sections 4 and 5 of the script from the top (the Stan section needs `drought` and `compare` from earlier sections; running the whole script takes about 2 min including compiles). First compile takes about a minute; sampling takes about a second.

Random effects as priors (~3 min)

- Same model as Monday. Hierarchical = multilevel = mixed. In Bayesian terms, a random effect is a parameter whose prior has estimated parameters.
- Four bullets: no integral; uncertainty carries through; priors help with few groups (callback to Monday's singular fit); the price.

Partial pooling (~10 min)

- Three choices on the board: complete, none, partial. Write w_j and read it as signal variance over (signal + sampling) variance.
- Code check: the formula matches `lmer()` to three decimals. $\hat{\sigma}_b = 0.113$, $\hat{\sigma} = 0.229$, $\hat{\mu} = 0.861$; weights 0.88 (30 yr) down to 0.55 (5 yr).
- Widget (`partial_pooling.html`): start at the defaults. Then $\sigma_b = 0$: every stand sits on the dashed line (complete pooling; both weight cards 0.00). Then $\sigma_b = 0.4$: weights 0.99 and 0.94, nearly no pooling. Then raise σ to 0.6 with σ_b back at 0.114: short records collapse toward the mean first.
- "Set σ_b to 0, then drag it to 0.4. Which two pooling choices are these? Then compare S10 and S12 ... Which moves more toward the overall mean, and why?" *Land:* complete pooling, then (nearly) no pooling. S12 (5 yr, $w = 0.55$) moves a larger share of its distance (45% vs. 34% for S10, $w = 0.66$). The absolute moves are almost equal (0.080 vs. 0.078 mm) because S10 starts farther from the mean; shrinkage is proportional to distance.
- Tie to the Normal-Normal posterior: the population is the prior.

Priors for a covariance matrix (~8 min)

- Board: $G = \text{diag}(\tau)\,\Omega\,\text{diag}(\tau)$; separate priors on τ and Ω. For 2×2, LKJ gives $p(\rho) \propto (1 - \rho^2)^{\eta - 1}$.
- Widget (`lkj_prior.html`), 2×2: $\eta = 1$ flat ($P(|\rho| > 0.5) = 0.50$); $\eta = 2$ (the Stan model's prior) 0.31; $\eta = 0.5$ U-shaped, 0.67. Ask students to predict 10×10 at $\eta = 1$ before switching.
- "For a 10×10 correlation matrix with $\eta = 1$, how likely is any one correlation to exceed 0.5 in absolute value? What does 'uniform' mean for a matrix that large?" *Land:* about 0.10 (prior SD 0.30). LKJ(1) is uniform over all valid correlation matrices, and most valid big matrices have small correlations (large ones would break positive definiteness), so each single correlation is pulled toward 0.
- One sentence on inverse-Wishart (old BUGS/JAGS default) and why LKJ replaced it.

Stan walkthrough (~25 min; laptops)

- Walk the blocks top to bottom. Two ideas to spend time on: the Cholesky factor (always a valid Ω; callback to Monday's third "harder" point) and non-centering (callback to the Week 9 funnel). `ab[stand, 1]` is multi-indexing: one row of `ab` per observation.
- Students compile and sample. No divergences; all $\hat{R} = 1.00$.
- Table: everything matches `lmer()` closely except ρ: MLE -0.57; posterior median -0.37, 90% interval -0.76 to 0.17; truth -0.6.
- Shrinkage figure: S11 (5 yr) moves most; S01 and S02 (30 yr) barely move. Don't claim the arrows follow the correlation (they mostly head toward the center).
- Truth check: in this data set the short-record stands' slopes barely improve (0.057 to 0.055) and their intercepts get worse (0.044 to 0.053). Over 500 simulated data sets (seed 746, `lmer()`), slope error falls 0.070 to 0.051 and intercept error 0.044 to 0.041. Shrinkage wins on average, not every time.
- "Why is the posterior for ρ so much wider ... closer to 0 than the MLE? Which summary is more honest?" *Land:* ρ is a correlation among 12 stand-level pairs, each

estimated with error; the MLE plugs in one value. The LKJ(2) prior pulls toward 0, and integrating over uncertainty in the τs spreads the posterior. The posterior (with its interval) is more honest; the MLE alone overstates what 12 stands say.

- "Change to `lkj_corr_cholesky(1)`, then `(10)` ..." *Land:* $\eta = 1$: median -0.45 (90% -0.81 to 0.12); $\eta = 10$: -0.13 (-0.45 to 0.21). The prior matters a lot for ρ because the data say little, but stand slopes move by at most 0.008 ($\eta = 1$) and 0.026 ($\eta = 10$) mm per 10 cm. Have half the room run each and compare.

brms (~7 min)

- One formula, same priors: cor -0.33 (95% -0.81 to 0.30), SDs 0.16 and 0.11. First compile ~20 s.
- Have students run `stancode(fit_brms)` and find `lkj_corr_cholesky`, `diag_pre_multiply`, and `z_1` in it (the same non-centered form).
- Defaults without `prior`: LKJ(1), half-Student-$t(3,\ 0,\ 2.5)$ on SDs, flat on slopes. `prior_summary()` shows them.

Discussion (~5 min; Q3 leads into the work session)

- Q1 (12 more stands or 20 more years?) *Land:* more stands. ρ and the τs describe the population of stands, and there are only 12 draws of (a_j, b_j). Longer records sharpen each stand's estimate but can't add stands; past a point they stop helping.
- Q2 (brms vs. your own Stan) *Land:* brms for anything a formula can say (fast, well-tested, sensible parameterizations); your own Stan for custom process models (nonlinear functions, latent states, observation models) or when you need control over every prior.
- Q3 (your project's groups) *Land:* open; push each student to name the grouping factor, how many levels it has, and whether slopes plausibly vary.

If time: ridge as a prior (~6 min) (planned for Week 10, never done). The six board steps and the `ridge_prior.html` walkthrough are in `NOTES_Week10_WebPageGuide.pdf` ("Ridge is a prior"); the widget is on the Week 10 page (`#bayesian`). Close with the page's bonus point: a random effect is a ridge penalty on the stand deviations, with τ estimated from the data instead of chosen by cross-validation.

Then: final-project work session for the rest of the lab period.